

mercial industries, agriculture, biomedical, cosmetics, environmental, food, water, and various scientific research fields. The electronic nose detects hazardous or poisonous gas, which is not possible for human sniffers.

Hands with a sense of touch

Electronic hands have to accomplish the human hands' most important ability—the sense of touch. Current prosthetic hands do little beyond filling the arm-shaped space between the body and air. People with prosthetics can't detect if they're in contact with an object without looking at it directly. A research team has solved this problem by inventing hands that send electric signals to brain.

When outfitted with prosthetic hands that stimulate the brain that way, monkeys responded just as though they physically touched objects themselves. Programming these same signals into artificial human limbs would give amputees replacement hands, unlike anything developed before.

Bionic arm and legs

Artificial arms and legs, or prosthetics, are intended to restore a degree of normal function to amputees. Prosthetic arms have been around since ancient Greece, but a fully mechanised one—controlled entirely by the mind—wasn't developed until recently.

Jesse Sullivan was the recipient of the world's first brain-controlled bionic arms, installed through nerve transfer surgery and managed by a computer embedded in his forearm. New plastics and other materials, such as carbon fibre, have allowed artificial limbs to become stronger and lighter, limiting the amount of extra energy necessary to operate the limb and to look much more realistic.

Artificial pancreas

An artificial pancreas is used to substitute the endocrine functionality of a healthy pancreas for diabetic and other patients who require it. It can be used for insulin replacement therapy



Fig. 8: Hand with a sense of touch



Fig. 9: A bionic leg

until glycemic control is practically normal as evidenced by the avoidance of complications of hyperglycemia. It can also ease the burden of therapy for insulin-dependent.

Artificial pancreas knocks insulin into the body automatically. The device looks much like a regular insulin pump, which slips insulin continuously through the skin. But this one monitors blood sugar at all times and adjusts itself accordingly. Unlike several electronic items for the human body, the artificial pancreas is still in early development stages.

Electronic bladder

Electrical stimulation is used to treat urinary incontinence by sending a mild electric current to nerves in the lower back or pelvic muscles that are involved in urination. The stimulation may make muscles contract, producing an effect that strengthens muscles by contracting these frequently. The

stimulation may also encourage the growth of nerve cells that causes muscles to contract.

Electrical stimulation of the bladder can also be done by placing electrodes under the skin, either through legs or into the lower back. The two main methods for replacing the bladder function involve either redirecting urine flow or replacing the bladder in situ. The standard method for replacing the bladder involves forming a bladder-like pouch from intestinal tissue. Methods to grow bladders using stem cells have been attempted in clinical research.

Electronic liver

Scientists are developing a bio-artificial liver device intended for the treatment of liver failure using stem cells. The artificial liver is designed to serve as a supportive device, either allowing the liver to regenerate upon failure or to bridge the patient's liver functions until a transplant is available. Though it uses real liver cells, it is not a permanent substitute.

Researchers have found that a mixture of human liver precursor cells (differentiated from human induced pluripotent stem cells) and two other cell types can spontaneously form three-dimensional structures dubbed 'liver buds.'

Electronic lungs

Artificial lungs promise to be a great success in the near future. A company is currently working on this type of medical device.

Extracorporeal membrane oxygenation (ECMO) can be used to take a significant load off of the native lung tissue and heart. In ECMO, one or more catheters are placed into a patient's body, and a pump is used to flow blood over hollow membrane fibres, which exchange oxygen and carbon dioxide with the blood.

Similar to ECMO, Extracorporeal CO₂ Removal (ECCO₂R) has a similar set-up but mainly benefits a patient through carbon dioxide removal rather than oxygenation, allowing the lungs to relax and heal. **EFY**